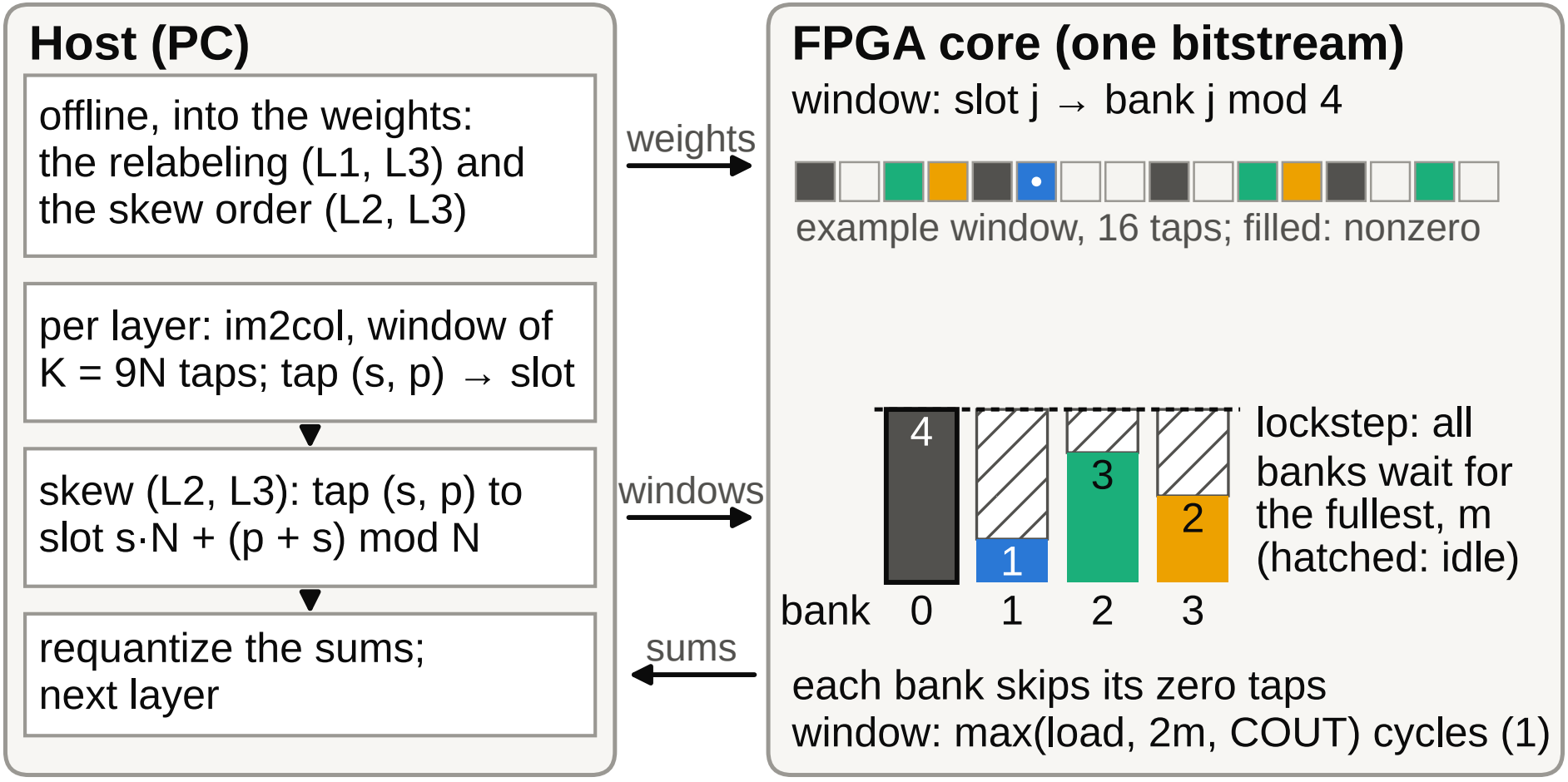
Host (PC)
offline, into the weights:
the relabeling (L1, L3) and
the skew order (L2, L3)
per layer: im2col, window of
K = 9N taps; tap (s, p) → slot
skew (L2, L3): tap (s, p) to
slot s·N + (p + s) mod N
requantize the sums;
next layer
weights
windows
sums
FPGA core (one bitstream)
window: slot j → bank j mod 4
example window, 16 taps; filled: nonzero
4
1
3
2
bank 0 1 2 3
lockstep: all
banks wait for
the fullest, m
(hatched: idle)
each bank skips its zero taps
window: max(load, 2m, COUT) cycles (1)